Tutorial 9

9.2 A current source in a linear circuit has

$$i_s = 15\cos(25\pi t + 25^\circ)\,A$$

a) What is the amplitude of the current: 15 A

b) What is the angular frequency? $\omega = 25\pi = 78.54\,rad/s$

c) Find the frequency: $f = \frac{\omega}{2\pi} = 12.5\,Hz$

d) Calculate i_s at 2ms:

$$I_s = 15\angle 25\,A$$

$$I_s(2ms) = 15\cos(25\pi)(2\times 10^{-3}) + 25^\circ)$$

$$= 15\cos(\tfrac{1}{20}\pi + 25^\circ)$$

$$= 15\cos(25.15^\circ)$$

$$= 13.58\,A$$

9. -or the following pairs of sinusoids, determine which one leads and by how much:

a) $v(t) = 10\cos(4t - 60^\circ)$

$i(t) = 4\sin(4t + 50^\circ) = 4\cos(4t + 50^\circ - 90^\circ)$

$= 4\cos(4t - 40^\circ)$

$\therefore$ $i(t)$ leads $v(t)$ by 20°

b) $v_1(t) = 4\cos(377t + 10^\circ)$

$V_2(t) = -20\cos 377t = 20\cos(377t + 180^\circ)$

$\therefore$ V_2 leads by 170°)

c) $x(t) = 13\cos 2t + 5\sin 2t = 13\cos 2t + 5\cos(2t - 90^\circ)$

$= 13\angle 0^\circ + 5\angle -90^\circ$

$$= 13 - j5 \quad = 13.928\angle -21.04^\circ$$

$$= 13.928\cos(2t - 21.04^\circ)$$

$y(t) = 15\cos(2t - 11.8^\circ)$

$\therefore$ $y(t)$ leads by $[-11.8 + 21.04 = 9.24^\circ]$

9.8 Calculate these complex numbers and express your results in rectangular form:

a) $$\frac{60\angle 45^\circ}{7.5 - j10} + j2 = \frac{60\angle 45^\circ}{12.5\angle -53.13} + j2$$

$$= 4.8\angle 98.13 + j2$$

$$= -0.6788 + j4.751 \quad + j2$$

$$= -0.6788 + j6.75$$

b) $$\frac{32\angle -20^\circ}{(6-j8)(4+j2)} + \frac{20}{-10+j24}$$

$\therefore (6-j8)(4+j2) = 24 + j12 - j32 + 16$

$$= 40 - j20 = 20\sqrt{5}\angle -26.56$$

$$\therefore \frac{32\angle -20^\circ}{20\sqrt{5}\angle -26.56} + \frac{20(-10-j24)}{(-10+j24)(-10-j24)}$$

$$= \frac{32\angle -20^\circ}{20\sqrt{5}\angle -26.56} + \frac{-200 - j480}{100 + 576}$$

$$= 0.711 + j0.0817 - 0.2958 - j0.71$$

$$= 0.415 - j0.6283$$

c) $20 + (16\angle -50^\circ)(5 + j12)$

$= 20 + (16\angle -50^\circ)(13\angle 67.38)$

$= 20 + 208\angle 17.38$

$= 218.5 + j62.13$

9.13 Evaluate the following complex numbers:

a) $\dfrac{2+j3}{1-j6} + \dfrac{7-j8}{-5+j11}$

$= (-0.4324 + j0.4054) + (-0.8425 - j0.2534)$

$= -1.2749 + j0.1520$

b) $\dfrac{(5\angle 10^\circ)(10\angle -40^\circ)}{(4\angle -80)(-6\angle 50^\circ)}$

$= \dfrac{50\angle -30^\circ}{24\angle 150^\circ} = -2.083$

c) $\begin{vmatrix} 2+j3 & -j2 \\ -j2 & 8-j5 \end{vmatrix}$

$= (2+j3)(8-j5) - (-4) = 35 + j14$

9.17 Two voltages v_1 and v_2 appear in series so that their sum is $v = v_1 + v_2$. If $v_1 = 10\cos(50t - \pi/3)$ V and $v_2 = 12\cos(50t + 30^\circ)$, find v:

$V = V_1 + V_2 = 10\angle -60^\circ + 12\angle 30^\circ$

$= 5 - j8.66 + 10.392 + j6$

$= 15.62\angle -9.805^\circ$

$\therefore v = 15.62\cos(50t - 9.8^\circ)$

9.19 Using phasors find:

a) $3\cos(20t+10^\circ) - 5\cos(20t-30^\circ)$

$= 3\angle 10^\circ - 5\angle -30^\circ = 2.954 + j0.5209 - 4.33 + j2.5$

$= -1.376 + j3.021$

$= 3.32\angle 114.49^\circ$

$\therefore \quad = 3.32\cos(20t + 114.49^\circ)$

b) $40\sin 50t + 30\cos(50t - 45^\circ)$

$= 40\angle -90^\circ + 30\angle -45^\circ = -j40 + 21.21 - j21.21$

$= 21.21 - j61.21$

$= 64.78\angle -70.89^\circ$

$= 64.78\cos(50t - 70.89^\circ)$

c) $20\sin 400t + 10\cos(400t + 60^\circ) - 5\sin(400t - 20^\circ)$

$\Rightarrow \sin\alpha = \cos(\alpha - 90^\circ)$

$= 20\angle -90^\circ + 10\angle 60^\circ - 5\angle -110^\circ = -j20 + 5 + j8.66 + 1.7101 + j4.69$

$= 6.7101 - j6.641$

$= 9.44\angle -44.7^\circ$

$= 9.44\cos(400t - 44.7^\circ)$

9.20 A linear network has a current input: $7.5\cos(10t + 30^\circ)$A and a voltage output $120\cos(10t + 75^\circ)$V. Determine impedance:

$I = 7.5\angle 30^\circ \quad V = 120\angle 75^\circ$

$$\therefore Z = \frac{V}{I} = \frac{120\angle 75}{7.5\angle 30} = 16\angle 45^\circ$$